

Angle, Orthogonal, and Orthonormal

- Using C-S inequality,

$$-1 \leq \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|} \leq 1$$

- Then, there exists a unique $\omega \in [0, \pi]$ with

$$\cos \omega = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

- We define ω as the **angle** between $\mathbf{x}$ and $\mathbf{y}$.

- Definition.** If $\langle \mathbf{x}, \mathbf{y} \rangle = 0$, in other words their angle is $\pi/2$, we say that they are **orthogonal**, denoted by $\mathbf{x} \perp \mathbf{y}$. Additionally, if $\|\mathbf{x}\| = \|\mathbf{y}\| = 1$, they are **orthonormal**.

Cosine similarity

$$\text{cs}(\vec{x}, \vec{y}) = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|} = \frac{\sum_1^n x_i y_i}{\sqrt{\sum_1^n x_i^2} \sqrt{\sum_1^n y_i^2}}$$

Example: $\vec{x} = [4, 2, 1, 2]^T$

$$\vec{y} = [1, 2, 2, 0]^T$$

$$\text{cs}(\vec{x}, \vec{y})$$

$$\vec{u} = 2\vec{x} = [8, 4, 2, 4]^T$$

$$\vec{v} = 3\vec{y} = [3, 6, 6, 0]^T$$

$$\text{cs}(\vec{u}, \vec{v})$$

Cosine similarity

$$cs(\vec{x}, \vec{y}) = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|} = \frac{\sum_1^n x_i y_i}{\sqrt{\sum_1^n x_i^2} \sqrt{\sum_1^n y_i^2}}$$

Example: $\vec{x} = [4, 2, 1, 2]^T$

$$\vec{y} = [1, 2, 2, 0]^T$$

$$\vec{u} = 2\vec{x} = [8, 4, 2, 4]^T$$

$$\vec{v} = 3\vec{y} = [3, 6, 6, 0]^T$$

$$\begin{aligned} cs(\vec{x}, \vec{y}) &= \frac{4*1+2*2+1*2+2*0}{\sqrt{4^2+2^2+1^2+2^2} \sqrt{1^2+2^2+2^2+0}} \\ &= \frac{10}{\sqrt{25}\sqrt{9}} = \frac{10}{15} = 0.67 \end{aligned}$$

$$\begin{aligned} cs(\vec{u}, \vec{v}) &= \frac{8*3+4*6+2*6+4*0}{\sqrt{8^2+4^2+2^2+4^2} \sqrt{3^2+6^2+6^2+0}} \\ &= \frac{60}{\sqrt{100}\sqrt{81}} = \frac{60}{90} = 0.67 \end{aligned}$$

Cosine similarity

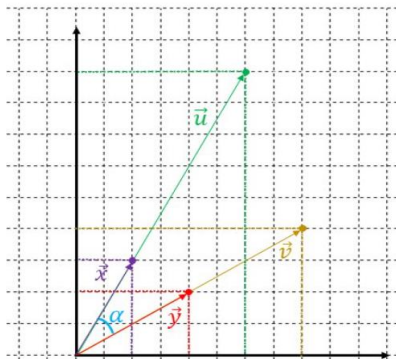
$$cs(\vec{x}, \vec{y}) = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|} = \frac{\sum_1^n x_i y_i}{\sqrt{\sum_1^n x_i^2} \sqrt{\sum_1^n y_i^2}}$$

$$\vec{x} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$\vec{u} = 3 * \vec{x} = \begin{pmatrix} 6 \\ 9 \end{pmatrix}$$

$$\vec{y} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$$

$$\vec{v} = 2 * \vec{y} = \begin{pmatrix} 8 \\ 4 \end{pmatrix}$$



Norm, Distance, Angle, Orthogonality

The definition of norm (or length), distance, angle, orthogonal, and normalizing for general inner product spaces closely parallel to those based on the dot product in Euclidean n -space

Norm (length) of $\mathbf{u}$: $\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$

Distance between $\mathbf{u}$ and $\mathbf{v}$: $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{\langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle}$

Angle between two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$:

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}, \quad 0 \leq \theta \leq \pi$$

Orthogonal: ($\mathbf{u} \perp \mathbf{v}$)

$\mathbf{u}$ and $\mathbf{v}$ are orthogonal if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$

Orthogonal, and Orthonormal

Example-Euclidean Space

- ▶ Orthogonal $\sim$ perpendicular.

- ▶ Orthogonal vectors x and y :

$$x^\top y = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n = 0$$

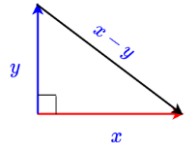
- Law of Cosines: θ is the angle between x and y :

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2\|x\|\|y\| \cos \theta$$

- Orthogonal vectors have $\cos \theta = 0$

Pythagoras Law:

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2$$

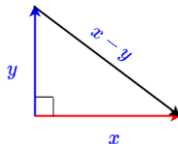


Orthogonal, and Orthonormal

Example-Euclidean Space

Pythagoras Law:

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2$$



Prove???

$$\begin{aligned}(x - y)^T (x - y) &= x^T x + y^T y \\ x^T x + y^T y - x^T y - y^T x &= x^T x + y^T y \\ x^T y &= 0\end{aligned}$$

Example

- Orthogonality is defined by a given inner product. Thus, different inner products may lead to different results about orthogonality.

- Example.** Consider two vectors $\mathbf{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{y} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{y}, \text{ they are not orthogonal.}$$

$$\cos \omega = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|} = -\frac{1}{3} \implies \omega \approx 1.91 \text{ rad} \approx 109.5^\circ$$

Orthogonal Matrix

- **Definition.** A square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is an **orthogonal matrix**, iff its columns (or rows) are **orthonormal** so that

$$\mathbf{A}\mathbf{A}^T = \mathbf{I} = \mathbf{A}^T\mathbf{A}, \text{ implying } \mathbf{A}^{-1} = \mathbf{A}^T.$$

- We can use $\mathbf{A}^{-1} = \mathbf{A}^T$ for the definition of orthogonal matrices.
- Fact 1. $\mathbf{A}, \mathbf{B}$: orthogonal $\implies \mathbf{AB}$: orthogonal
- Fact 2. $\mathbf{A}$: orthogonal $\implies \det(\mathbf{A}) = \pm 1$

Orthogonal Matrix

- The linear mapping Φ by orthogonal matrices preserve **length** and **angle** (for the dot product)

$$\begin{aligned}\|\Phi(\mathbf{A})\| &= \|\mathbf{Ax}\|^2 = (\mathbf{Ax})^T (\mathbf{Ax}) = \mathbf{x}^T \mathbf{A}^T \mathbf{Ax} = \mathbf{x}^T \mathbf{x} = \|\mathbf{x}\|^2 \\ \cos \omega &= \frac{(\mathbf{Ax})^T (\mathbf{Ay})}{\|\mathbf{Ax}\| \|\mathbf{Ay}\|} = \frac{\mathbf{x}^T \mathbf{A}^T \mathbf{Ay}}{\sqrt{\mathbf{x}^T \mathbf{A}^T \mathbf{Ax} \mathbf{y}^T \mathbf{A}^T \mathbf{Ay}}} = \frac{\mathbf{x}^T \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}\end{aligned}$$

Orthonormal Basis

- Basis that is orthonormal, i.e., they are all orthogonal to each other and their lengths are 1.
- Standard basis in $\mathbb{R}^n$, $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$, is orthonormal.
- **Question.** How to obtain an orthonormal basis?

1. Use Gaussian elimination to find a basis for a vector space spanned by a set of vectors.
 - Given a set $\{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ of unorthogonal and unnormalized basis vectors. Apply Gaussian elimination to the augmented matrix $(\mathbf{B}\mathbf{B}^T | \mathbf{B})$
2. Constructive way: Gram-Schmidt process (we will cover this later)

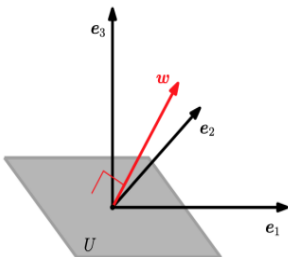
Orthogonal Complement

- Consider D -dimensional vector space V and M -dimensional subspace $W \subset V$. The **orthogonal complement** $U^\perp$ is a $(D - M)$ -dimensional subspace of V and contains all vectors in V that are orthogonal to every vector in U .
- $U \cap U^\perp = 0$
- Any vector $x \in V$ can be uniquely decomposed into:

$$\mathbf{x} = \sum_{m=1}^M \lambda_m \mathbf{b}_m + \sum_{j=1}^{D-M} \psi_j \mathbf{b}_j^\perp, \quad \lambda_m, \psi_j \in \mathbb{R},$$

where $(\mathbf{b}_1, \dots, \mathbf{b}_M)$ and $(\mathbf{b}_1^\perp, \dots, \mathbf{b}_{D-M}^\perp)$ are the **bases** of U and $U^\perp$, respectively.

Orthogonal Complement



- The vector $\mathbf{w}$ with $\|\mathbf{w}\| = 1$, which is orthogonal to U , is the basis of $U^\perp$.
- Such $\mathbf{w}$ is called **normal vector** to U .
- For a linear mapping represented by a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, the solution space of $\mathbf{A}\mathbf{x} = 0$ is $\text{row}(\mathbf{A})^\perp$, where $\text{row}(\mathbf{A})$ is the row space of $\mathbf{A}$ (i.e., span of row vectors)
In other words, $\text{row}(\mathbf{A})^\perp = \ker(\mathbf{A})$

Fundamental Theorem in Linear Algebra

The row space and nullspace of A are orthogonal complements in $\mathbb{R}^n$.

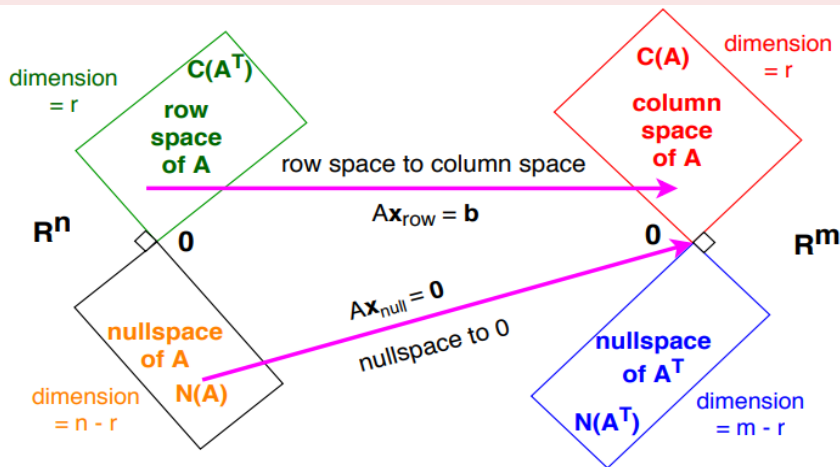
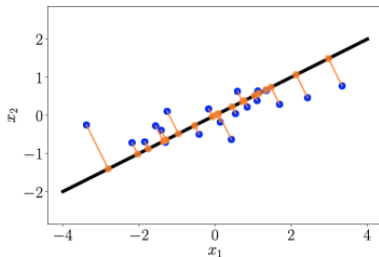


Figure: Two pairs of orthogonal subspaces.

Orthogonal Projections

- Big data: high dimensional
- However, most information is contained in a few dimensions
- **Projection**: A process of reducing the dimensions (hopefully) without loss of much information
- Example. Projection of 2D dataset onto 1D subspace



Dimensionality reduction

Many modern data domains involve huge numbers of features / dimensions

- + Documents: thousands of words, millions of bigrams
- + Images: thousands to millions of pixels
- + Genomics: thousands of genes, millions of DNA polymorphisms

Why reduce dimensions?

High dimensionality has many costs

- + Redundant and irrelevant features degrade performance of some ML algorithms
- + Difficulty in interpretation and visualization
- + Computation may become infeasible

Approaches to dimensionality reduction

- Feature selection
 - + Select subset of existing features (without modification)
- Model regularization
 - + L2 reduces *effective* dimensionality
 - + L1 reduces *actual* dimensionality
- Combine (map) existing features into smaller number of new features
 - + Linear combination (projection)
 - + Nonlinear combination

Linear dimensionality reduction

- Linearly project n -dimensional data onto a k -dimensional space
 - + $k < n$, often $k \ll n$
 - + Example: project space of 10^4 words into 3 dimensions
- There are infinitely many k -dimensional subspaces we can project the data onto.
- Which one should we choose?

Linear dimensionality reduction

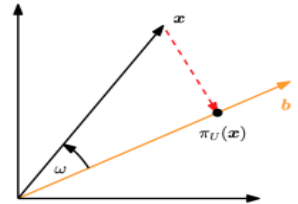
- Best k -dimensional subspace for projection depends on task
 - + Classification: maximize separation among classes
 - Example: linear discriminant analysis (LDA)
 - + Regression: maximize correlation between projected data and response variable
 - Example: partial least squares (PLS)
 - + Unsupervised: retain as much data variance as possible
 - Example: principal component analysis (PCA)

Projection onto Lines (1D Subspaces)

- Consider a 1D subspace $U \subset \mathbb{R}^n$ spanned by the basis $\mathbf{b}$.
- For $\mathbf{x} \in \mathbb{R}^n$, what is its projection $\pi_U(\mathbf{x})$ onto U (assume the dot product)?

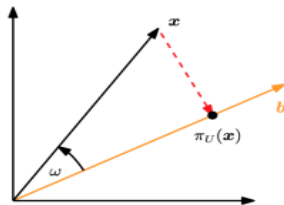
$$\langle \mathbf{x} - \pi_U(\mathbf{x}), \mathbf{b} \rangle = 0 \xleftrightarrow{\pi_U(\mathbf{x}) = \lambda \mathbf{b}} \langle \mathbf{x} - \lambda \mathbf{b}, \mathbf{b} \rangle = 0$$

$$\Rightarrow \lambda = \frac{\langle \mathbf{b}, \mathbf{x} \rangle}{\|\mathbf{b}\|^2} = \frac{\mathbf{b}^T \mathbf{x}}{\|\mathbf{b}\|^2}, \text{ and } \pi_U(\mathbf{x}) = \lambda \mathbf{b} = \frac{\mathbf{b}^T \mathbf{x}}{\|\mathbf{b}\|^2} \mathbf{b}$$



(a) Projection of $\mathbf{x} \in \mathbb{R}^2$ onto a subspace U with basis vector $\mathbf{b}$.

Projection onto Lines (1D Subspaces)



(a) Projection of $\mathbf{x} \in \mathbb{R}^2$ onto a subspace U with basis vector $\mathbf{b}$.

- Projection matrix $\mathbf{P}_\pi \in \mathbb{R}^{n \times n}$ in $\pi_U(\mathbf{x}) = \mathbf{P}_\pi \mathbf{x}$

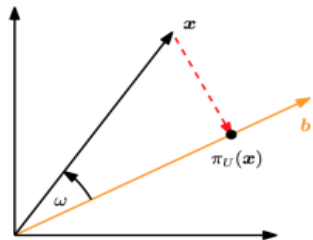
$$\pi_U(\mathbf{x}) = \lambda \mathbf{b} = \mathbf{b} \lambda = \frac{\mathbf{b} \mathbf{b}^\top}{\|\mathbf{b}\|^2} \mathbf{x}, \quad \mathbf{P}_\pi = \frac{\mathbf{b} \mathbf{b}^\top}{\|\mathbf{b}\|^2}$$

Inner Product and Projection

- We project $\mathbf{x}$ onto $\mathbf{b}$, and let $\pi_{\mathbf{b}}(\mathbf{x})$ be the projected vector.
- **Question.** Understanding the inner product $\langle \mathbf{x}, \mathbf{b} \rangle$ from the projection perspective?

$$\langle \mathbf{x}, \mathbf{b} \rangle = \|\pi_{\mathbf{b}}(\mathbf{x})\| \times \|\mathbf{b}\|$$

- In other words, the inner product of $\mathbf{x}$ and $\mathbf{b}$ is the product of (length of the projection of $\mathbf{x}$ onto $\mathbf{b}$) $\times$ (length of $\mathbf{b}$)



(a) Projection of $\mathbf{x} \in \mathbb{R}^2$ onto a subspace U with basis vector $\mathbf{b}$.

Example

$$\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$$

$$P_{\pi}$$

$$\mathbf{x} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\pi_U(\mathbf{x})$$

Câu 5 Tìm hình chiếu của vector $b = \begin{bmatrix} x \\ y \end{bmatrix}$ lên vector $a = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$? Nếu *chữ số cuối cùng trong MSSV của bạn* là số chẵn thì $b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, nếu là số lẻ thì $b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. (GI)

A. $\begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$	B. $\begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix}$	C. $\begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix}$	D. $\begin{bmatrix} -0.5 \\ -0.5 \end{bmatrix}$
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Câu 7 Tìm hình chiếu của vector $b = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}$ lên không gian cột của ma trận

$$Q = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \\ 2 & 2 \end{bmatrix} \text{? (GI)}$$

A. $\begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix}$	B. $\begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix}$	C. $\begin{bmatrix} 2/3 \\ 2/3 \\ 8/3 \end{bmatrix}$	D. $\begin{bmatrix} 8/3 \\ 2/3 \\ 2/3 \end{bmatrix}$
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Example

Example

$$\bullet \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$$

$$\mathbf{P}_\pi = \frac{\mathbf{b}\mathbf{b}^\top}{\|\mathbf{b}\|^2} = \frac{1}{9} \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} (1 \ 2 \ 2) = \frac{1}{9} \begin{pmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}$$

$$\text{For } \mathbf{x} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},$$

$$\pi_U(\mathbf{x}) = \mathbf{P}_\pi \mathbf{x} = \frac{1}{9} \begin{pmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{9} \begin{pmatrix} 5 \\ 10 \\ 10 \end{pmatrix} \in \text{span} \left[\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \right]$$

Project $b = (3, 3, 3)$ on the Q1 line

$$Q_1 = \frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$$

$$\frac{1}{9} \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} [2 \quad 2 \quad -1] \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} 9 = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$$

Projection onto General Subspace

- $\mathbb{R}^n \rightarrow 1\text{-Dim}$

- A basis vector $\mathbf{b}$ in 1D subspace

$$\pi_U(\mathbf{x}) = \frac{\mathbf{b}\mathbf{b}^\top \mathbf{x}}{\mathbf{b}^\top \mathbf{b}}, \quad \lambda = \frac{\mathbf{b}^\top \mathbf{x}}{\mathbf{b}^\top \mathbf{b}}$$

$$P_\pi = \frac{\mathbf{b}\mathbf{b}^\top}{\mathbf{b}^\top \mathbf{b}}$$

- $\mathbb{R}^n \rightarrow m\text{-Dim}, (m < n)$

- A basis matrix

$$B = (\mathbf{b}_1, \dots, \mathbf{b}_m) \in \mathbb{R}^{n \times m}$$

$$\pi_U(\mathbf{x}) = B(B^\top B)^{-1} B^\top \mathbf{x}, \quad \lambda = (B^\top B)^{-1} B^\top \mathbf{x}$$

$$P_\pi = B(B^\top B)^{-1} B^\top$$

- $\lambda \in \mathbb{R}^1$ and $\boldsymbol{\lambda} \in \mathbb{R}^m$ are the coordinates in the projected spaces, respectively.
- $(B^\top B)^{-1} B^\top$ is called **pseudo-inverse**.

Example

$$U = \text{span}\left[\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}\right] \subset \mathbb{R}^3 \text{ and } \mathbf{x} = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}$$

Compute $\pi_U(\mathbf{x})$

Check that $\{(1 \ 1 \ 1)^T, (0 \ 1 \ 2)^T\}$ is a basis.

Let $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$. Then, $\mathbf{B}^T \mathbf{B} = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}$

$$\mathbf{P}_\pi = \mathbf{B}(\mathbf{B}^T \mathbf{B})^{-1} \mathbf{B}^T = \frac{1}{6} \begin{pmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{pmatrix} \quad \pi_U(\mathbf{x}) = \frac{1}{6} \begin{pmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{pmatrix} \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}$$

ORTHOGONAL SETS

— Show that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthonormal basis of $\mathbb{R}^3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 3/\sqrt{11} \\ 1/\sqrt{11} \\ 1/\sqrt{11} \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} -1/\sqrt{66} \\ -4/\sqrt{66} \\ 7/\sqrt{66} \end{bmatrix}$$

Solution: Compute

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = -3/\sqrt{66} + 2/\sqrt{66} + 1/\sqrt{66} = 0$$

$$\mathbf{v}_1 \cdot \mathbf{v}_3 = -3/\sqrt{726} - 4/\sqrt{726} + 7/\sqrt{726} = 0$$

$$\mathbf{v}_2 \cdot \mathbf{v}_3 = 1/\sqrt{396} - 8/\sqrt{396} + 7/\sqrt{396} = 0$$

$$\vec{a} = \begin{bmatrix} 4 \\ 2 \\ -1 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix} \quad \text{these were orthogonal}$$

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{1}{\sqrt{21}} \begin{bmatrix} 4 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 4/\sqrt{21} \\ 2/\sqrt{21} \\ -1/\sqrt{21} \end{bmatrix}$$

← generated by normalization

$$\hat{b} = \frac{\vec{b}}{|\vec{b}|} = \frac{1}{\sqrt{14}} \begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{14} \\ -3/\sqrt{14} \\ -2/\sqrt{14} \end{bmatrix}$$

←

Gram–Schmidt algorithm

1. Let $B = \{v_1, v_2, \dots, v_n\}$ be a basis for an inner product space V .
2. Let $B' = \{w_1, w_2, \dots, w_n\}$, where

$$w_1 = v_1$$

$$w_2 = v_2 - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1$$

$$w_3 = v_3 - \frac{\langle v_3, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_3, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2$$

$$\vdots$$

$$w_n = v_n - \frac{\langle v_n, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_n, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2 - \dots - \frac{\langle v_n, w_{n-1} \rangle}{\langle w_{n-1}, w_{n-1} \rangle} w_{n-1}.$$

Then B' is an *orthogonal* basis for V .

3. Let $u_i = \frac{w_i}{\|w_i\|}$. Then $B'' = \{u_1, u_2, \dots, u_n\}$ is an *orthonormal* basis for V .

Example

$$\vec{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \vec{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$\vec{u}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\vec{u}_2 = \vec{v}_2 - \frac{\vec{v}_2 \cdot \vec{u}_1}{\vec{u}_1 \cdot \vec{u}_1} \vec{u}_1 = \begin{bmatrix} 1/3 \\ 2/3 \\ 1/3 \end{bmatrix}$$

$$\vec{u}_3 = \vec{v}_3 - \frac{\vec{v}_3 \cdot \vec{u}_1}{\vec{u}_1 \cdot \vec{u}_1} \vec{u}_1 - \frac{\vec{v}_3 \cdot \vec{u}_2}{\vec{u}_2 \cdot \vec{u}_2} \vec{u}_2 = \begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix}$$

Example

$$\vec{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \vec{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$\vec{u}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \Rightarrow \sqrt{\vec{u}_1 \cdot \vec{u}_1} = \sqrt{3} \quad \frac{\vec{u}_1}{\sqrt{3}} = \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

$$\vec{u}_2 = \begin{bmatrix} 1/3 \\ 2/3 \\ 1/3 \end{bmatrix} \Rightarrow \sqrt{\vec{u}_2 \cdot \vec{u}_2} = \sqrt{2/3} \quad \frac{\vec{u}_2}{\sqrt{2/3}} = \begin{bmatrix} 1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}$$

$$\vec{u}_3 = \begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix} \Rightarrow \sqrt{\vec{u}_3 \cdot \vec{u}_3} = \sqrt{1/2} \quad \frac{\vec{u}_3}{\sqrt{1/2}} = \begin{bmatrix} -\sqrt{2}/2 \\ 0 \\ \sqrt{2}/2 \end{bmatrix}$$

Positive/Negative Definite Matrix

Definitions:

- 1) An $n \times n$ symmetric real matrix $\mathbf{A}$ is said to be **positive-definite** if $x^T \mathbf{A} x > 0$ for all non-zero $x \in \mathbb{R}^n$
- 2) An $n \times n$ symmetric real matrix $\mathbf{A}$ is said to be **positive-semidefinite** if $x^T \mathbf{A} x \geq 0$ for all non-zero $x \in \mathbb{R}^n$
- 3) An $n \times n$ symmetric real matrix $\mathbf{A}$ is said to be **negative-definite** if $x^T \mathbf{A} x < 0$ for all non-zero $x \in \mathbb{R}^n$
- 4) An $n \times n$ symmetric real matrix $\mathbf{A}$ is said to be **negative-semi definite** if $x^T \mathbf{A} x \leq 0$ for all non-zero $x \in \mathbb{R}^n$

Example 1: $A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

$$\begin{aligned} x^T A x &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2x_1 \\ x_2 \end{bmatrix} \\ &= 2x_1^2 + x_2^2 \end{aligned}$$

whenever $x_1 \neq 0$ or $x_2 \neq 0$ (hence $x \neq 0$), the matrix A is positive definite.

Example 2: $A = \begin{bmatrix} 9 & -15 \\ -15 & 25 \end{bmatrix}$

$$\begin{aligned} x^T A x &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 9 & -15 \\ -15 & 25 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 9x_1 - 15x_2 \\ -15x_1 + 25x_2 \end{bmatrix} \\ &= 9x_1^2 - 15x_1x_2 - 15x_1x_2 + 25x_2^2 \\ &= (3x_1 - 5x_2)^2 \end{aligned}$$

Then, $x^T A x \geq 0$ if $x \neq 0$ and A is positive semi-definite. However, it is not positive definite because there exist non-zero vectors, for example the vector

$$x = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

Positive/Negative Definite Matrix

Definitions:

- 1) An $n \times n$ symmetric real matrix $\mathbf{A}$ is said to be **positive-definite** if $x^T \mathbf{A} x > 0$ for all non-zero $x \in \mathbb{R}^n$
- 2) An $n \times n$ symmetric real matrix $\mathbf{A}$ is said to be **positive-semidefinite** if $x^T \mathbf{A} x \geq 0$ for all non-zero $x \in \mathbb{R}^n$
- 3) An $n \times n$ symmetric real matrix $\mathbf{A}$ is said to be **negative-definite** if $x^T \mathbf{A} x < 0$ for all non-zero $x \in \mathbb{R}^n$
- 4) An $n \times n$ symmetric real matrix $\mathbf{A}$ is said to be **negative-definite** if $x^T \mathbf{A} x \leq 0$ for all non-zero $x \in \mathbb{R}^n$

Theory:

Let $\mathbf{A}$ be $n \times n$ symmetric real matrix $\mathbf{A}$. All eigenvalues of $\mathbf{A}$ are real.

- 1) $\mathbf{A}$ is positive definite if and only if all of its eigenvalues are positive
- 2) $\mathbf{A}$ is positive semi-definite if and only if all of its eigenvalues are non-negative.
- 3) $\mathbf{A}$ is negative definite if and only if all of its eigenvalues are negative
- 4) $\mathbf{A}$ is negative semi-definite if and only if all of its eigenvalues are non-positive.
- 5) $\mathbf{A}$ is indefinite if and only if it has both positive and negative eigenvalues.

Eigenvalue and Eigenvector

Definition

Let A be an $n \times n$ matrix

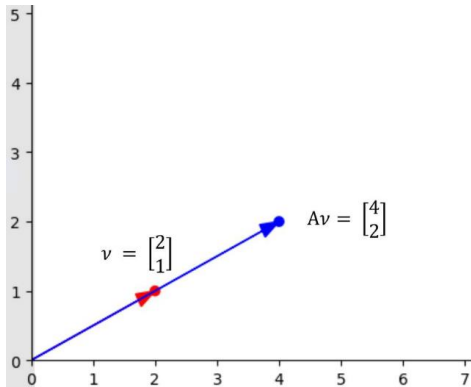
1. An ***eigenvector*** of A is a *nonzero* vector v in $\mathbf{R}^n$ such that $Av = \lambda v$, for some scalar λ
2. An ***eigenvalue*** of A is a scalar λ such that the equation $Av = \lambda v$ has a *nontrivial* solution

If $Av = \lambda v$ for $v \neq 0$, we say that λ is the ***eigenvalue for*** v , and that v is an ***eigenvector for*** λ

Note. Eigenvalues and eigenvectors are only for square matrices

Eigenvalue and Eigenvector

Definition



$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$

$$v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

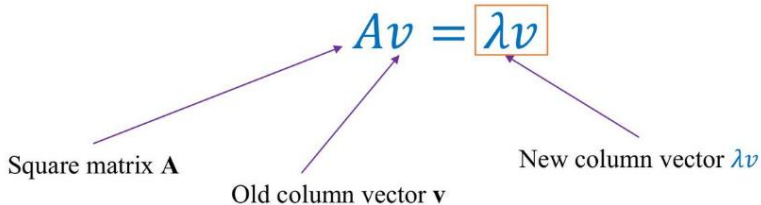
$$Av = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \lambda v$$

$$\lambda v = \lambda \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\lambda = 2$$

Eigenvalue and Eigenvector

Definition



Vector v is call eigenvector of matrix A

If we multiply Matrix A by vector v , the new vector λv does not change direction after the transformation

Eigenvalue and Eigenvector

Procedure and example

$$A = \begin{pmatrix} -6 & 3 \\ 4 & 5 \end{pmatrix}$$

$$Av = \lambda v$$

$$\Leftrightarrow Av - \lambda v = 0$$

$$\Leftrightarrow Av - \lambda Iv = 0$$

$$\Leftrightarrow (A - \lambda I)v = 0$$

If v is non-zero then we can solve for λ using just the **determinant**

$$|A - \lambda I| = 0$$

Start with $|A - \lambda I| = 0$

$$\left| \begin{pmatrix} -6 & 3 \\ 4 & 5 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right| = 0$$

Which is

$$\begin{vmatrix} -6 - \lambda & 3 \\ 4 & 5 - \lambda \end{vmatrix} = 0$$

Calculating that determinant gets:

$$(-6 - \lambda)(5 - \lambda) - 3 \times 4 = 0$$

Which then gets us this

$$\lambda^2 + \lambda - 42 = 0$$

And solving it gets:

$$\lambda = -7 \text{ or } \lambda = 6$$

Eigenvalue and Eigenvector

Procedure and example

Start with

$$Av = \lambda v$$

Put in the values we know

$$\begin{pmatrix} -6 & 3 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 6 \begin{pmatrix} x \\ y \end{pmatrix}$$

After multiplying we get these two equations:

$$\begin{aligned} -6x + 3y &= 6x \\ 4x + 5y &= 6y \end{aligned}$$

Bringing all to left hand side:

$$\begin{aligned} -12x + 3y &= 0 \\ 4x - 1y &= 0 \end{aligned}$$

Either equation reveals that $y = 4x$, so the *eigenvector* is a *nonzero* multiple of this:

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

And we get the solution:

$$\begin{pmatrix} -6 & 3 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} -6 \times 1 + 3 \times 4 \\ 4 \times 1 + 5 \times 4 \end{pmatrix} = \begin{pmatrix} 6 \\ 24 \end{pmatrix}$$

... and also...

$$6 \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 6 \\ 24 \end{pmatrix}$$

So $Av = \lambda v$

Positive definiteness

- **Test 1:** A matrix **A** will be positive definite if all its eigenvalues are positive; that is, all the values of λ that satisfy the determinantal equation

$$|\mathbf{A} - \lambda \mathbf{I}| = 0$$

should be positive.

Negative definiteness

- Equivalently, a matrix is **negative-definite** if all its **eigenvalues** are **negative**
- It is **positive-semidefinite** if all its **eigenvalues** are all greater than or equal to zero
- It is **negative-semidefinite** if all its **eigenvalues** are all less than or equal to zero

Example:

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$\text{Det}(A - \lambda I_3) = \begin{vmatrix} 2 - \lambda & 1 & 0 \\ 1 & 2 - \lambda & 0 \\ 0 & 0 & 3 - \lambda \end{vmatrix}$$

$$= (3 - \lambda) \begin{vmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix}$$

$$= (1 - \lambda)(3 - \lambda)^2$$

$$\text{Det}(A - \lambda I_3) = 0 \Leftrightarrow \lambda = 1 \text{ or } \lambda = 3$$

All the Eigen Values are positive. The matrix is positive-definite.

Example:

$$A = \begin{pmatrix} 6 & 5 & 12 \\ 5 & 19 & 0 \\ 12 & 3 & 7 \end{pmatrix}$$

$$\text{Det}(A - \lambda I_3) = \begin{vmatrix} 6 - \lambda & 5 & 12 \\ 5 & 19 - \lambda & 0 \\ 12 & 3 & 7 - \lambda \end{vmatrix} = -(\lambda^3 - 32\lambda^2 + 111\lambda + 1807)$$

Eigen Values obtained: $\lambda = -5.611$; $\lambda = 24.42$; $\lambda = 13.18$

Not all the Eigen Values are positive.
 So, the matrix is Not positive-definite.

Positive definiteness

- **Test 2:** Another test that can be used to find the positive definiteness of a matrix **A** of order n involves evaluation of the determinants

$$A = |a_{11}|$$

$$A_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

$$A_3 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$A_n = \begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & & & & \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix}$$

- The matrix **A** will be **positive definite** if and only if all the values $A_1, A_2, A_3, \dots, A_n$ are positive
- The matrix **A** will be **negative definite** if and only if the sign of A_j is $(-1)^j$ for $j=1, 2, \dots, n$
- If some of the A_j are positive and the remaining A_j are zero, the matrix **A** will be **positive semidefinite**

Example:

$$A = \begin{pmatrix} 6 & 5 & 12 \\ 5 & 19 & 0 \\ 12 & 3 & 7 \end{pmatrix}$$

A???

Solution: Compute Det

$$A_1 = (6); A_2 = \begin{pmatrix} 6 & 5 \\ 5 & 9 \end{pmatrix}; A_3 = \begin{pmatrix} 6 & 5 & 12 \\ 5 & 19 & 0 \\ 12 & 3 & 7 \end{pmatrix}$$